TIME-OF-FLIGHT DISTANCE SENSOR

ARK DIST

DroneCAN and MAVLink laser distance sensor in short-, mid- and long-range variants

EXPOSED INTERFACES CAN UART SWD

The USA-built ARK DIST pairs a Broadcom AFBR-S50 time-of-flight sensor with an STM32F412VG running the PX4 DroneCAN node firmware. The SR, MR and LR variants share one board and case and differ only in the sensor module.



01 KEY FEATURES

- Broadcom AFBR-S50 time-of-flight sensor, 850 nm Class 1 laser
- Max reported distance: SR 30 m, MR 50 m, LR 100 m
- DroneCAN and MAVLink 2 output, firmware update over CAN
- 2× CAN and UART on Pixhawk-standard JST-GH, debug on JST-SH
- PX4 and ArduPilot; ROS 2 via ros2_dronecan

02 SPECIFICATIONS

SENSOR	
Time-of-flight module	Broadcom AFBR-S50, see 03
Laser	850 nm, IEC 60825-1 Class 1 (eye-safe)
Transmitter beam	2° × 2°, 3.5 cm spot at 1 m
Receiver (SR, MR)	32 pixels, 12.4° × 6.2° field of view

INTERFACES	
DroneCAN	uavcan.equipment.range_sensor.Measurement
CAN functions	Firmware update, dynamic node allocation
UART	MAVLink 2 DISTANCE_SENSOR and HEARTBEAT
UART settings	115200 8N1, RTS/CTS optional (auto-detected)
MAVLink system / component ID	158 / 158
Debug	NSH console and SWD

PROCESSOR & FIRMWARE	
MCU	STM32F412VG
Firmware target	ark_dist_default, board ID 92
First release with all variants	ark_dist-v1.18.0.1

ELECTRICAL & MECHANICAL	
Supply voltage	5 V
Supply current	See 03
Dimensions	20 × 28 × 14 mm
Weight	4 g

AT A GLANCE	
Sensor	Broadcom AFBR-S50
Variants	SR / MR / LR
Max reported distance	30 / 50 / 100 m
Update rate	20 / 15 / 10 Hz
Supply	5 V
Dimensions	20 × 28 × 14 mm
Weight	4 g
Hardware rev	Rev 1.0

COMPLIANCE & SOURCING

- Made in the USA
- NDAA compliant
- IEC 60825-1 Class 1 laser product

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-dist-sr/
arkelectron.com/product/ark-dist-mr/
arkelectron.com/product/ark-dist-lr/

DOCS docs.arkelectron.com/products/sensor/ark-dist

CAD github.com/ARK-Electronics/ark-hardware

03 VARIANTS

	ARK DIST SR	ARK DIST MR	ARK DIST LR
Time-of-flight module	AFBR-S50LV85D	AFBR-S50LX85D	AFBR-S50EX85D
Measurement mode	Long Range, DFM 4X	Long Range, DFM 4X	Extended Long Range
Update rate	20 Hz (Fast profile 50 Hz)	15 Hz (Fast profile 25 Hz)	10 Hz
Max reported distance	30 m	50 m	100 m
Min reported distance	0 m	0.1 m	0.6 m
Supply current, avg / max	84 / 86 mA	78 / 84 mA	TBD · not yet measured

Mode, rate and reported-distance limits are shipped firmware defaults. The Fast profile is SENS_AFBR_PROF = 1.

04 MEASURED RANGE

SURFACE	ARK DIST SR		ARK DIST MR		ARK DIST LR	
	BRIGHT SUN	DUSK	BRIGHT SUN	DUSK	BRIGHT SUN	DUSK
Asphalt	18 m	—	16 m	—	18 m	40 m
Grass	—	30 m*	—	50 m*	40 m	82 m

Measured reliable range: the farthest distance with ≥50 % of readings valid and within 5 m of the reference, flown at default settings, not derated. Bright sun: sun 49–61° above the horizon; a lower sun gives more range. Dusk: sun below 6°. * Limited by the maximum reported distance; the sensor was still measuring at the flight ceiling (SR 38 m, MR 54 m). — Not measured. Wet surfaces reduce range; open water (~2 % reflectivity) gives only short, intermittent returns.

05 CAN CONNECTOR (×2)

PIN	SIGNAL	VOLTAGE
1	5V	5.0 V
2	CAN_P	5.0 V
3	CAN_N	5.0 V
4	GND	GND

Connector: 4-pin JST-GH, Pixhawk standard. Chain additional nodes from the second CAN connector.

06 UART CONNECTOR

PIN	SIGNAL	VOLTAGE
1	5V_UART	5.0 V
2	USART3_RX	3.3 V
3	USART3_TX	3.3 V
4	USART3_RTS	3.3 V
5	USART3_CTS	3.3 V
6	GND	GND

Connector: 6-pin JST-GH, Pixhawk standard.

07 DEBUG CONNECTOR

PIN	SIGNAL	VOLTAGE
1	3.3V	3.3 V
2	USART2_TX	3.3 V
3	USART2_RX	3.3 V
4	FMU_SWDIO	3.3 V
5	FMU_SWCLK	3.3 V
6	GND	GND

Connector: 6-pin JST-SH, Pixhawk standard.

08 LED INDICATORS

STATUS LED	MEANING
Green, fast blink (10 Hz)	Node firmware running. It does not change with CAN traffic.
Green, slow blink (1 Hz)	Bootloader listening for CAN traffic
Blue, blinking (2 Hz)	Bootloader waiting for a node ID from the flight controller
Blue and green, blinking (3 Hz)	Bootloader waiting up to 3 s for the flight controller to start a firmware update
Red, green and blue together (3 Hz)	Firmware update in progress
Red, blinking	Firmware update error. 1 Hz: the flight controller returned a file error. 2 Hz: the flight controller stopped responding. 4 Hz: the image failed its CRC check. A failed update restarts the node after 20 s.

Without valid firmware the node stays in the bootloader until the flight controller flashes it.

ARK-DS-ARK-DIST | Issue 2026-10-12 | Compiled from ARK Electronics published documentation, the ark_dist-v1.18.0.1 firmware defaults, and ARK flight measurements. Laser and optical characteristics per Broadcom. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.